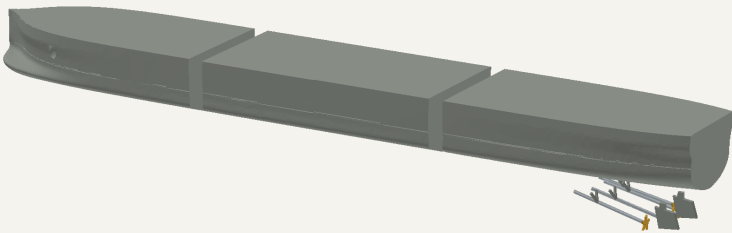


Hull Assembly Guide

USS DWIGHT D. EISENHOWER (CVN-69) · MUSEUM EDITION · V0.1.0

MILESTONE 1 — EXPLODED HULL ASSEMBLY

THREE KEYED HULL MODULES · FOUR-SHAFT RUNNING GEAR · TWIN RUDDERS



Assembly inventory

Part group	Qty	Material / color	Locator
Hull modules	3	PLA Matte Ash Gray	Asymmetric paired internal keys
Shafts	4	PLA Silk Silver	Concealed hex root grooves
A-bracket struts	8	PLA Matte Ash Gray	Concealed hull-end sockets
Propellers	4	PLA Basic Gold	Blind shaft-tip bore
Rudders	2	PLA Matte Ash Gray	Concealed rectangular stock

No screws, magnets, rods, metal parts, inserts, or electronics are required. Total printed-part count: 21.

1. Join the hull modules

- Deburr only the hidden keys and sockets. Preserve the exterior seam faces and engraved waterline.
- Dry-fit bow module 1 to midship module 2. The asymmetric keys prevent inversion. The seam must close by hand without force.
- Wet only the female sockets with a small amount of PLA-compatible adhesive. Join on a flat surface and confirm the top interface is coplanar.
- After full cure, repeat for midship module 2 and stern module 3.

2. Install the running gear

Line	Position	Shaft	Propeller
1	Port outer	Shaft_1	Propeller_1
2	Port inner	Shaft_2	Propeller_2
3	Starboard inner	Shaft_3	Propeller_3
4	Starboard outer	Shaft_4	Propeller_4

- Seat each numbered shaft in its concealed groove without adhesive; confirm four distinct, symmetric centerlines.
- Fit the matching P/S A-bracket struts. Glue the hull end first, then wick a minimal amount to the shaft contact.
- Slide each matching propeller onto the shaft tip. The blind bore controls concentricity and axial position.
- Insert the port and starboard rudder stocks upward into their stern slots. Set both rudders parallel before cure.

3. Final acceptance

- Top interface remains flat across both module seams.
- Exterior surfaces show no adhesive bloom or squeeze-out.
- Shaft lines, propeller disks, and rudders are symmetric in plan and elevation.
- Model is supported in a padded cradle; propellers and rudders never carry display load.

First-article caveat

The 0.25 mm-per-side fit allowance passed geometric interference checks. Printer calibration, filament moisture, and shrinkage still require a physical dry fit before adhesive is applied.